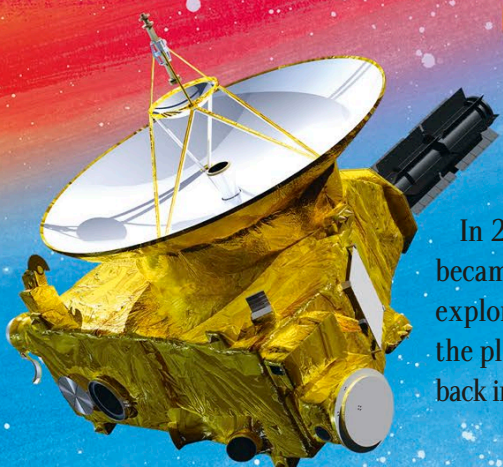
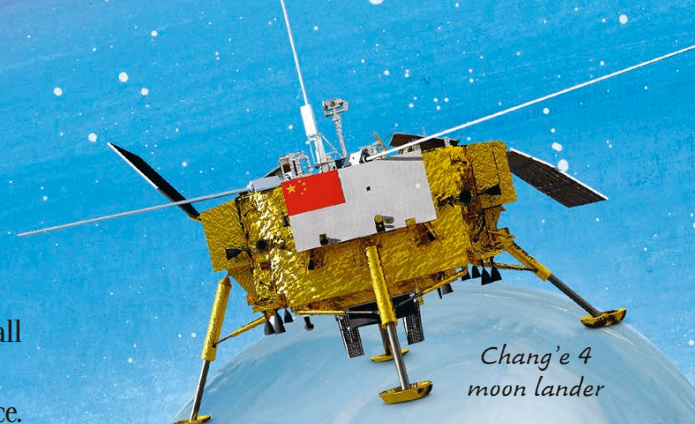


A gold-colored spacecraft with a large white parabolic dish antenna and various instruments, set against a background of a red and orange planet and a blue sky with stars.

New Horizons

In 2015, the New Horizons probe became the first spacecraft to closely explore Pluto, a dwarf planet beyond all the planets in the solar system. It sent back incredible photographs of the surface.

A gold-colored lunar lander with a red flag, standing on a grey spherical moon. A large blue Earth is in the background.

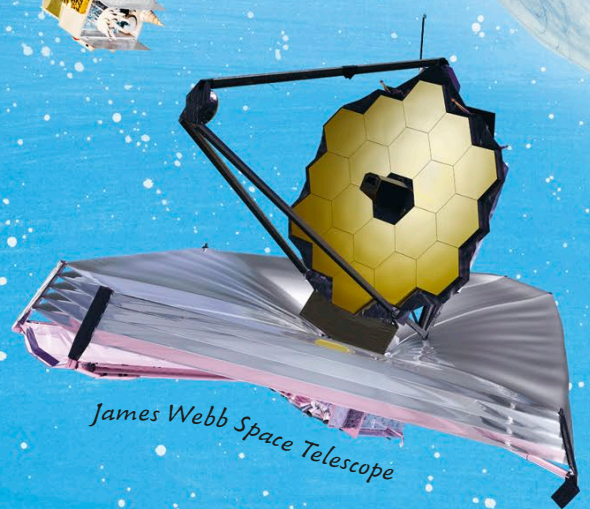
Moon lander

Landers are robots designed to land on the surface of planets, moons, asteroids, and comets. They are packed with instruments that only work when they are close to an object. This lunar lander touched down on the moon in 2019.

A satellite with a central body and four large rectangular solar panel arrays, floating in space.

Weather satellite

A huge network of satellites monitors the Earth's weather and climate from positions high above the planet. They use radar, cameras, and other tools to do this.

A large space telescope with a gold-colored hexagonal mirror and a complex support structure, with a pinkish solar sail or instrument cover.

Seismometer

Seismometers detect vibrations of the ground, including movements too small for us to feel. They can tell us about earthquakes and volcanic explosions, but also provide clues about what the Earth is like deep underground.

A cross-section of two volcanoes with orange lava flows. In the foreground, a seismometer is shown on a tripod stand next to a small drum.

What can't be measured?

One problem in science is that we can only observe things that our senses or instruments can detect. This adds up to less than 5 percent of the universe! The stuff we can't detect yet is known as dark matter and dark energy.

A circular view through a space telescope showing a colorful nebula with stars, set against a dark blue background with stars.

Space telescope

Telescopes placed in orbit around the Earth get a better view of space. The light and other radiation they collect hasn't had to travel through the Earth's atmosphere first.